

ORIGINAL ARTICLE

Testicular torsion–detorsion causes dysfunction of mitochondrial oxidative phosphorylation

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Abstract

Background: Semen quality impairment is a serious consequence of testicular torsion–detorsion. Adequate germ-cell mitochondrial oxidative phosphorylation plays a crucial role in male fertility. Changes in cellular oxidative phosphorylation in testicular tissues after testicular torsion–detorsion remain unclear.

Objectives: This study investigated whether testicular torsion–detorsion induces alternations of mitochondrial oxidative phosphorylation in testicular tissues.

Materials and Methods: BALB/c male mice were divided into a Sham group and a testicular torsion–detorsion group. At the end of the procedure, the mice were euthanized, and their bilateral testicles were removed. Mitochondria morphology was evaluated through transmission electron microscopy. The cellular respiratory functions of germ cells were evaluated using a Seahorse analyzer assay. The proteome profiles in testicular tissues were analyzed using liquid chromatography–tandem mass spectrometry. The differences in the expression levels of each component in the oxidative phosphorylation were revealed using Ingenuity Pathways Analysis.

Results: Inner mitochondrial membrane disruption was found in ipsilateral twisted testicular mitochondria in the torsion–detorsion group but not in contralateral untwisted testes. The cellular respiratory function in germ cells was significantly decreased after testicular torsion–detorsion in ipsilateral twisted testes but not in contralateral untwisted testes. Liquid chromatography–tandem mass spectrometry analysis

of ipsilateral twisted testicular tissue revealed that mitochondrial proteins were differentially expressed after testicular torsion–detorsion. Testicular torsion–detorsion induced downregulation of oxidative phosphorylation and revealed alternations of specific proteins in the oxidative phosphorylation complexes.

Discussion and Conclusion: Testicular torsion–detorsion produced mitochondria injury and dysregulation of mitochondrial oxidative phosphorylation in ipsilateral twisted testes. Different protein expressions were identified in the mitochondrial oxidative phosphorylation complexes with testicular torsion–detorsion; new therapeutic targets may be identified to restore the oxidative phosphorylation function of germ cells.

KEYWORDS

oxidative, phosphorylation, reperfusion, testis, torsion

1 | INTRODUCTION

Testicular torsion caused by a twisted spermatic cord induces testicular ischemia injury. The detorsion of the testes induces further injury through reperfusion injury. In animal studies, testicular torsion–detorsion (TD) was noted to increase oxidative and inflammatory responses in testicular tissues and decrease spermatogenesis in germ cells because of apoptotic injury.^{1,2} In retrospective human studies, certain patients experienced impaired spermatogenesis, including decreased sperm count and motility, after testicular TD, even after detorsion of the ischemic testis.^{3–5} Because sperm concentration and motility are critical factors for successful fertilization,^{6,7} improvement of sperm quality is a crucial for patients with testicular TD.

The sufficiency of adenosine triphosphate (ATP) production by functional mitochondria in germ cells is the basis for efficient spermatogenesis and good semen quality.^{8,9} ATP production in sperm cells is dependent on oxidative phosphorylation (OXPHOS).¹⁰ Mitochondrial dysfunction, such as defects in OXPHOS complexes, is correlated with sperm functional parameters and impaired fertilization capacity.^{10–12} Altering the expression of components in OXPHOS complexes has been reported to induce poor sperm quality.^{8,13} Reperfusion injury–induced dysregulation of cellular bioenergetics plays a major role in organ dysfunction in the heart, liver, and brain.^{14–16} Previous studies have reported that testicular TD reduces sperm ATP production.¹⁷ However, the relationship between testicular TD and the OXPHOS of germ cells remains unclear. To evaluate the status of cellular OXPHOS in germ cells after testicular TD, this study used a testicular TD model in mice to evaluate the impact of testicular TD on OXPHOS in testicular tissues. Furthermore, no effective treatment for preventing testicular injury induced by reperfusion injury is currently clinically available. One unmet need for patients with testicular TD is the identification of dysregulated proteins in testicular tissues after testicular TD. We used liquid chromatography–tandem mass spectrometry (LC–MS/MS) and Ingenuity Pathway Analysis (IPA) to identify dysregulated proteins in OXPHOS complexes after testicular TD.

2 | MATERIALS AND METHODS

2.1 | Animals

This study used young adult male BALB/c mice (22–26 g; BioLASCO, Taipei, Taiwan). This animal study was approved by the Institutional Animal Care and Use Committee of Taipei Medical University (LAC-2018-0074). The care and handling of the mice were conducted in accordance with the guidelines of the National Institutes of Health, USA.

2.2 | Testicular TD model

Testicular TD was performed per a previously described method.² In brief, a mixture of zoletil/xylazine (40/10 mg/kg) was intraperitoneally injected to anesthetize the mice, after which a scrotal midline incision was performed. Testicular torsion was created through a 720° clockwise rotation of the left testis, followed by fixation to the scrotum. All mice remained anesthetized for the following 2 h with supplementary intraperitoneal doses of zoletil/xylazine (40/10 mg/kg per 30 min). In the detorsion group, the left testis was untwisted to its natural position. The sham operation included similar surgical procedures except that the left testis remained untwisted. Bilateral testes were removed after 24 h of detorsion or sham procedure. The mice were divided into the ipsilateral testis of Sham group (testes receive sham procedure), contralateral testis of Sham group (testes do not receive any procedure), ipsilateral testis of TD group (testes receive TD procedure), and contralateral testis of TD group (testes do not receive any procedure) ($n = 6$ in each group).

2.3 | Histopathology

Testicular tissues were sectioned and then stained using hematoxylin and eosin. The testicular tissues were evaluated microscopically by a

pathologist blinded to the experimental groups. The injury score of testicular tissues was according to a previous report.² Each characteristic (hemorrhage, edema, vascular congestion, polymorphonuclear leukocyte infiltration, and germ cell ischemia) was scored on a scale from 0 = *normal* to 3 = *severe*. Overall testicular injury of each specimen was evaluated according to the total scores (0–5, normal to mild injury; 6–10, moderate injury; 11–15, severe injury).

2.4 | Evaluation of mitochondrial morphology through transmission electron microscopy

The testicular tissues were processed as previously described.¹⁸ The sections of bilateral testes in each group were examined using a transmission electron microscope (Hitachi HT-7700; Hitachi, Ltd., Tokyo, Japan). The morphologies of mitochondria in testicular tissues and spermatozoa were evaluated; as previously described, mitochondrial structural deformity was defined as being either deformation of the mitochondrial cristae or inner mitochondrial membrane disruption.¹⁹ Five random fields ($2\ \mu\text{m}^2$) were selected for counting the number of deformed mitochondria.

2.5 | LC-MS/MS analysis

Protein solutions (10 μg) of testicular tissues in ipsilateral testes of TD and Sham groups were diluted and digested with sequencing-grade modified porcine trypsin (Promega, Madison, WI, USA) at 37°C for 16 h. Peptides were desalted and dried through vacuum centrifugation, and they were stored at a temperature of –80°C until use. The digested peptides were diluted in high-performance liquid chromatography (HPLC) buffer A (0.1% formic acid; Sigma–Aldrich, St. Louis, MO, USA), loaded into a reversed-phase pre-column (Zorbax 300SB-C18, $0.3 \times 5\ \text{mm}^2$; Agilent Technologies, Wilmington, DE, USA), and then separated on a homemade column (HydroRP 2.5 μm , 75 μm I.D. $\times$ 20 cm with a 15- μm tip) using a multistep gradient of HPLC buffer B (99.9% acetonitrile/0.1% formic acid) for 70 min. The liquid chromatography apparatus (Dionex RSLC system; Thermo Fisher Scientific, San Jose, CA, USA) with electrospray interface was coupled with Orbitrap Elite™ Hybrid Ion Trap-Orbitrap Mass Spectrometer (Thermo Fisher Scientific) operated using Xcalibur 2.2 software (Thermo Fisher Scientific). Mass spectrometry (MS) was operated in data-dependent acquisition mode of which internal calibration was performed followed by one full MS scan and 20 tandem MS (MS/MS) scan events for the 20 most abundant precursor ions in the full MS scan. The automatic gain control was set as maximum accumulated time of 100 ms or ion counts of 3×10^6 ions for full MS scan and 200 ms or 3×10^3 ions for MS/MS scans.

2.6 | Protein identification and quantitation

Protein identification and quantitation was performed using Proteome Discoverer software (version 2.2; Thermo Fisher Scientific). The MS/MS spectra were searched against the SwissProt database

(extracted from *Mus musculus*, 16,992 sequences) using the Mascot (Matrix Science, London, UK; version 2.5). For peptide identification, mass tolerances of 10 ppm and 0.5 Da were set for precursor and fragment ions, respectively. Tryptic peptides with up to two missed cleavages were allowed. Only high-confidence ($p < 0.05$) and ranked one peptide–spectrum matches were filtered for confident peptide identification with an overall false discovery rate < 0.01 . Proteins with single peptide hits were removed.

To estimate comparative protein abundance in the testis tissues, label-free semiquantitation of protein levels was determined by using spectral counting method.²⁰ The normalized spectral count of each protein in the sample was calculated by dividing the spectral count of a given protein by the total spectral counts of the sample, and then multiplying the average total spectral count of each sample. The fold change of normalized spectral count of each protein in TD group relative to Sham group was recorded.

2.7 | Functional enrichment analysis by using IPA

The differentially expressed proteins with a fold change higher or lower than 1.5 in TD group compared to Sham group were selected for functional enrichment analysis by using IPA (Qiagen Inc., Valencia, CA, USA). The enriched canonical pathways in the ipsilateral testes of TD and Sham groups were verified using the right-tailed Fisher's exact test and presented as the negative logarithm of the unadjusted p -value ($-\log(p)$). The pathway was considered significant if the $-\log(p) > 1.3$ ($p < 0.05$). The z -score represented the activation states (upregulated or downregulated) of canonical pathways.

2.8 | Immunoblotting assay

For validation of the downregulated proteins in OXPHOS complexes in ipsilateral testicular tissues, immunoblotting assay was performed. Snap frozen testicular tissues were thawed, homogenized, microwaved, and centrifuged and then the supernatants were collected. Protein concentration of each sample was measured (bicinchoninic acid protein assay kit; Pierce Chemical Co., Rockford, IL, USA). Equal amount of proteins was separated by electrophoresis and then transferred to nitrocellulose membranes (Bio-Rad Laboratories, Hercules, CA, USA). Then, the membranes were incubated overnight with primary antibody of NUDFS1, ATP5C1, SDHC, ATP5J, or actin (as the internal standard; Sigma–Aldrich). Bound antibody was detected by chemiluminescence (ECL plus kit; Amersham BioSciences, Buckinghamshire, UK). The images were acquired by multispectral imaging system (UVP BioSpectrum 815 Imaging System) and analyzed by Vision Works LS software (UVP, California, USA).

2.9 | Isolation of germ cells

Germ cells were isolated from the bilateral testes of TD and Sham groups following the protocol of Chang et al.²¹ Specifically, and in

brief, the testicular tunica albuginea was decapsulated, and the seminiferous tubules were dissociated. Seminiferous tubules were dissolved through enzymatic digestion and countercurrent centrifugal elutriation. Leydig and Sertoli cells were discarded, and germ cells were collected. The purity of the germ cells was determined using immunofluorescence staining with the germ-cell-specific marker VASA (Figure S1).² In brief, the isolated cells were incubated overnight at 4°C with the primary antibody of VASA (Boster, Pleasanton, CA, USA) and then treated with a fluorescent rhodamine isothiocyanate-conjugated secondary antibody (Jackson ImmunoResearch, West Grove, PA, USA) for 1 h to detect germ cells. The sections stained with immunofluorescence were evaluated using a confocal microscope (LMS-780; Carl Zeiss, Oberkochen, Germany).

2.10 | Evaluation of the respiratory functions of germ cells

The cellular metabolic activity of isolated germ cells in each group was evaluated by Cell Counting Kit-8 (CCK-8) assay following corresponding instructions. Briefly, isolated germ cells were seeded at 1.5×10^4 /well and cultured for 48 h. CCK-8 kit (10 μ l; Sigma-Aldrich) was added and the incubation was sustained for 4 h. The absorbance intensity was determined at 450 nm by a microplate reader (Molecular Devices, San Jose, CA, USA). Mean values of the absorbance intensities in each group (ipsilateral and contralateral testes) were calculated. All experiments were performed in triplicate and cell relative viability was expressed as a percentage relative to the Sham group.

The isolated germ cells were cultured in culture medium for 48 h before performing a Seahorse assay in an incubator at 32°C with 5% carbon dioxide. The culture medium was Dulbecco's minimal essential medium, containing 10% fetal bovine serum with 1 mM sodium pyruvate, 2 mM L-glutamine, 12 ng/ml mouse leukemia inhibitory factor, 20 ng/ml bovine basic fibroblast growth factor, 0.04 ng/ml human interleukin-11, 10 ng/ml human stem cell factor, 10 ng/ml oncostatin M, 100 units/ml penicillin, and 100 mg/ml streptomycin. Before Seahorse assay, counting of viable cells with Trypan blue staining was performed. The germ cells were then transferred to a 24-well cell culture microplate (10,000 cells/well) with Seahorse XF assay medium. The mitochondrial function of the germ cells was evaluated using the Seahorse XF Analyzer, per a previously described method.²² The optimal concentration of carbonyl cyanide p-trifluoromethoxyphenyl hydrazine (FCCP) was determined by using different FCCP concentrations with the Seahorse XF Cell Mito Stress Test Kit. The respiratory function of germ cells was measured by recording the oxygen consumption rate (OCR) at the basal condition and after adding mitochondrial respiration modulators (oligomycin, FCCP, rotenone, and antimycin) using the Seahorse XF Cell Mito Stress Test Kit (Agilent Technologies, Santa Clara, CA). The level of mitochondrial ATP production can be recorded based on the reduction in basal OCR in response to oligomycin. The maximum OCR after adding FCCP is the maximum rate of respiration that the germ cells can achieve.

2.11 | Statistical analysis

Between-group differences were analyzed using an unpaired t-test. Statistical significance was indicated if $p < 0.05$. The *prism* package in GraphPad (GraphPad Software, Inc., La Jolla, CA, USA) was used for statistical analyses.

3 | RESULTS

3.1 | Testicular TD-induced testicular injury and disruption of testicular mitochondrial morphology

Histological findings showed moderate testicular injury in the ipsilateral testis of TD group (mean injury score 10) and normal to mild injury in the ipsilateral testis of Sham group (mean injury score 1.8) (Figure S2). The injury score was significantly higher in TD group than Sham group in ipsilateral testicular tissues ($p < 0.001$; Figure S2). In contralateral testicular tissues, normal to mild testicular injury was noted in Sham (mean injury score 1) and TD (mean injury score 1.8) groups (Figure S2).

In the ipsilateral testicular tissues, normal mitochondrial morphology was observed in the Sham group (Figure 1A), and both deformation of the mitochondrial cristae and inner mitochondrial membrane disruption were noted in the TD group (Figure 1A). The number of disrupted mitochondria was significantly higher in the TD group than in the Sham group ($p < 0.001$; Figure 1B). In the contralateral testes, the numbers of intact and disrupted mitochondria in testicular tissue were not significantly difference between Sham and TD group ($p = 0.1014$; Figure 1A). These results indicated that detorsion of the ischemic testis caused mitochondrial damage in the twisted ipsilateral testes but not in the untwisted contralateral testes.

An intact mitochondrial morphology was preserved in the mid-piece region of the sperm cells in the ipsilateral testis of Sham group (Figure 1C). In the midpiece region of the sperm cells in the ipsilateral twisted testicular tissues, disruption of the inner mitochondrial membrane was noted (Figure 1C). In the ipsilateral testes, the number of disrupted mitochondria was significantly higher in the TD group than in the Sham group ($p < 0.001$; Figure 1D). In the contralateral testes, the numbers of intact and disrupted mitochondria did not differ significantly between the groups ($p = 0.132$; Figure 1D). These results demonstrated that testicular TD induced mitochondrial morphology disruption in sperm cells in the twisted ipsilateral testes; however, mitochondrial morphology remained unchanged in the untwisted contralateral testes.

3.2 | Identification of proteins in twisted ipsilateral testes through LC-MS/MS

Mitochondrial morphology disruption was observed in the ipsilateral testes; identifying the affected canonical pathways and specific proteins after testicular detorsion is important to prevent the further testicular injury. The protein profiles of the ipsilateral testes in the

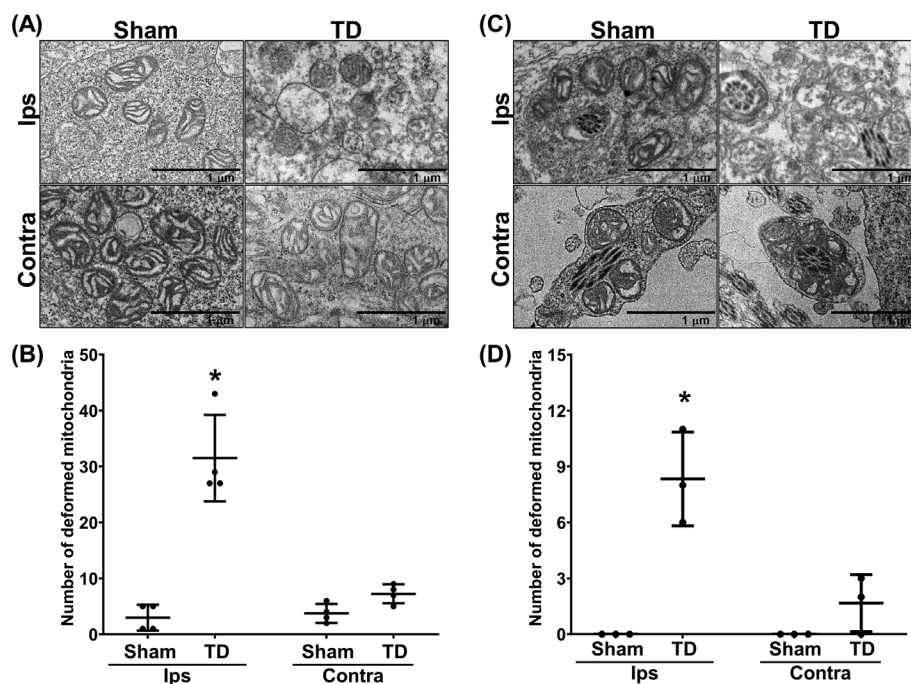


FIGURE 1 Representative transmission electron microscope images of mitochondria. Morphology of mitochondria in testicular tissue (A) and sperm (C). Number of deformed mitochondria in testicular tissues (B) and spermatozoa (D). Data regarding morphology of mitochondria in testicular tissue and spermatozoa were obtained from four and three mice in each group, respectively. TD, torsion-detorsion; Ips, ipsilateral twisted testis; Contra, contralateral untwisted testis. * $p < 0.05$ versus Sham group. Scale bar: $1 \mu\text{m}$

Sham and TD groups were analyzed using LC-MS/MS. The average numbers of identified proteins in the twisted ipsilateral testes were 873 in the Sham group and 871 in the TD group. For mitochondria-related proteins, 89 (Sham group) and 84 (TD group) proteins were identified in the ipsilateral testes. The proteins with a fold change more than 1.5 ($n = 40$) or less than 1.5 ($n = 52$) in TD group compared to Sham group were selected to pathway enrichment analysis (Figures S4 and S5).

3.3 | Dysregulation pathways related to OXPHOS induced by testicular TD

The canonical pathways in the ipsilateral testes were expressed differently between the groups. Figure 2 presents the differential canonical pathways between the TD and Sham groups. In the ipsilateral testis, the OXPHOS was the only downregulated pathway in the TD group compared to Sham group (Figure 2). These results revealed that testicular TD induced OXPHOS-related canonical pathways dysregulation in twisted ipsilateral testes.

3.4 | Testicular TD-induced cellular respiratory dysfunction in germ cells

The CCK-8 was used to evaluate the cellular metabolic activity of germ cells. In ipsilateral testes, the mean metabolic activity of germ cells in TD group was 52.41% compared to Sham group ($p = 0.001$; Figure S3). In contralateral testes, the mean metabolic activity of germ

cells in TD groups was 116.74% compared to Sham group ($p = 0.21$; Figure S3). These data revealed that detorsion of the twisted ipsilateral testes induced metabolic activity impairment in germ cells.

The mitochondrial respiratory function of germ cells isolated from testes was measured using the Seahorse XF Analyzer (Figure 3). The cell survival rates of germ cells before the Seahorse assay were $71.54\% \pm 4.5\%$ in the Sham group and $65.95\% \pm 5.53\%$ in the TD group ($p = 0.25$). In ipsilateral testes, the results revealed that basal respiratory rate was significantly lower in the TD group than in the Sham group ($p < 0.001$; Figure 3B). For ATP production ($p = 0.002$; Figure 3C), proton leaks ($p = 0.005$; Figure 3D), maximal respiratory function ($p = 0.007$; Figure 3E), and spare respiratory capacity ($p = 0.001$; Figure 3F), the OCR values were significantly lower in the TD group than in the Sham group. However, none of the parameters for respiratory functions in germ cells that were isolated from the contralateral testes differed significantly between the TD and Sham groups (Figure 3). These data revealed that detorsion of the twisted ipsilateral testes induced mitochondrial respiratory function impairment in germ cells. However, this phenomenon was not observed in the contralateral testes.

3.5 | Downregulation of NDUFS1, SDHC, ATP5J, and ATP5C1 results in mitochondrial respiratory dysfunction

After detorsion of the twisted ipsilateral testis, IPA analysis revealed a dysregulation of proteins involved in OXPHOS complexes in the TD

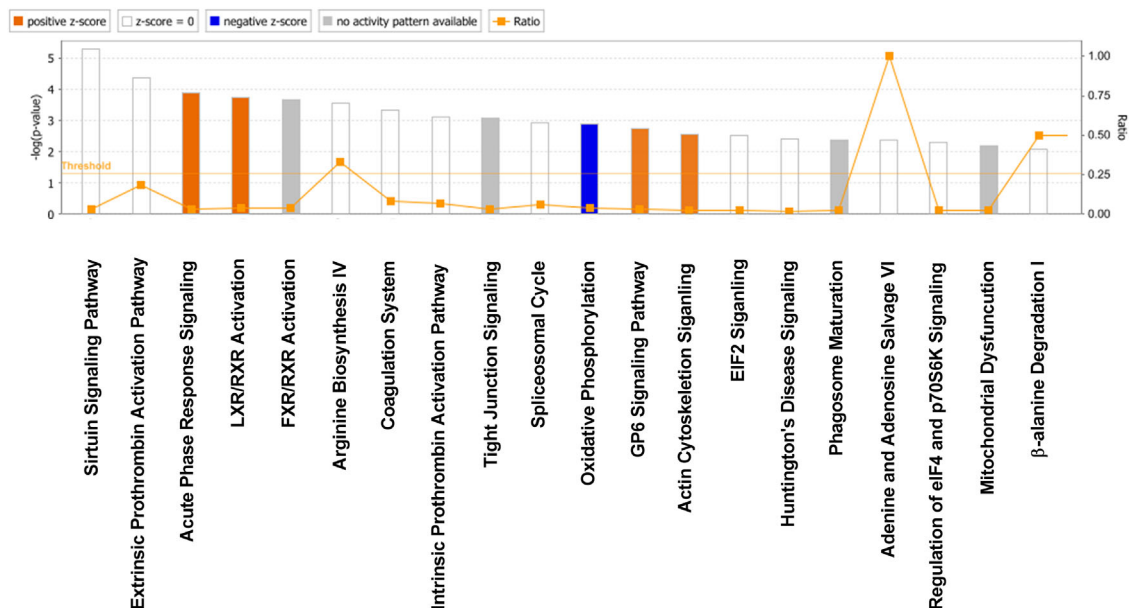


FIGURE 2 Representative top 20 canonical pathways identified using Ingenuity Pathway Analysis in torsion-detorsion (TD) group and Sham group. The horizontal axis represents the pathways, and the right vertical axis represents the negative logarithm of the unadjusted p -value ($-\log(p)$). The threshold of a significant canonical pathway is set to 1.3, with p -value ≤ 0.05 . The right vertical axis represents the ratio of the number of differentially expressed proteins in the testes to total number of proteins in each pathway. Orange bar: pathways are upregulated in the TD group. Blue bar: pathways are downregulated in the TD group. Gray bar: no prediction can be made

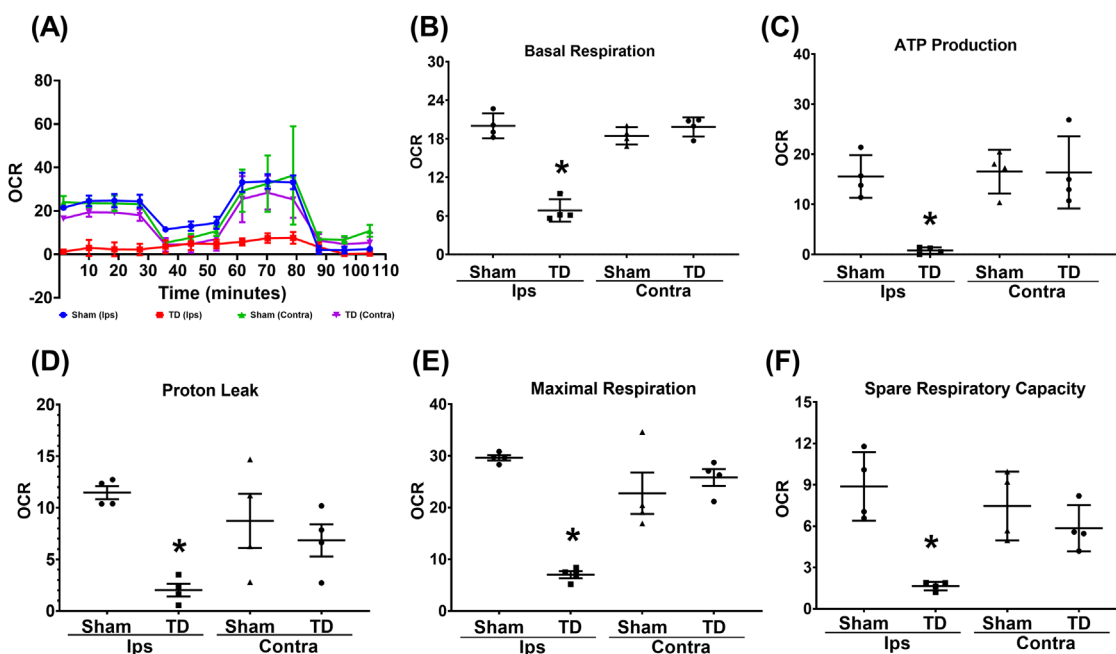


FIGURE 3 Representative results of Seahorse assay for mitochondrial respiratory function of germ cells. Data were derived from four mice in each group. OCR, oxygen consumption rate; TD, torsion-detorsion; ATP, adenosine triphosphate; Ips, ipsilateral twisted testis; Contra, contralateral untwisted testis. * $p < 0.05$ versus Sham group

group relative to the Sham group (Figure 4). These results demonstrated that testicular detorsion induced downregulation of complex I, II, and V of OXPHOS complexes in the twisted ipsilateral testes; testicular TD induced NADH:ubiquinone oxidoreductase core subunit S1 (NDUFS1) in complex I, succinate dehydrogenase complex subunit C

(SDHC) in complex II, ATP5J in complex V, and ATP5C1 in complex V (Figure 4B). These proteins may play an important role in OXPHOS dysfunction in testicular tissues after testicular TD. To further validate the impact of downregulated proteins in OXPHOS complexes after testicular TD, we applied western blot analysis for these downregulated

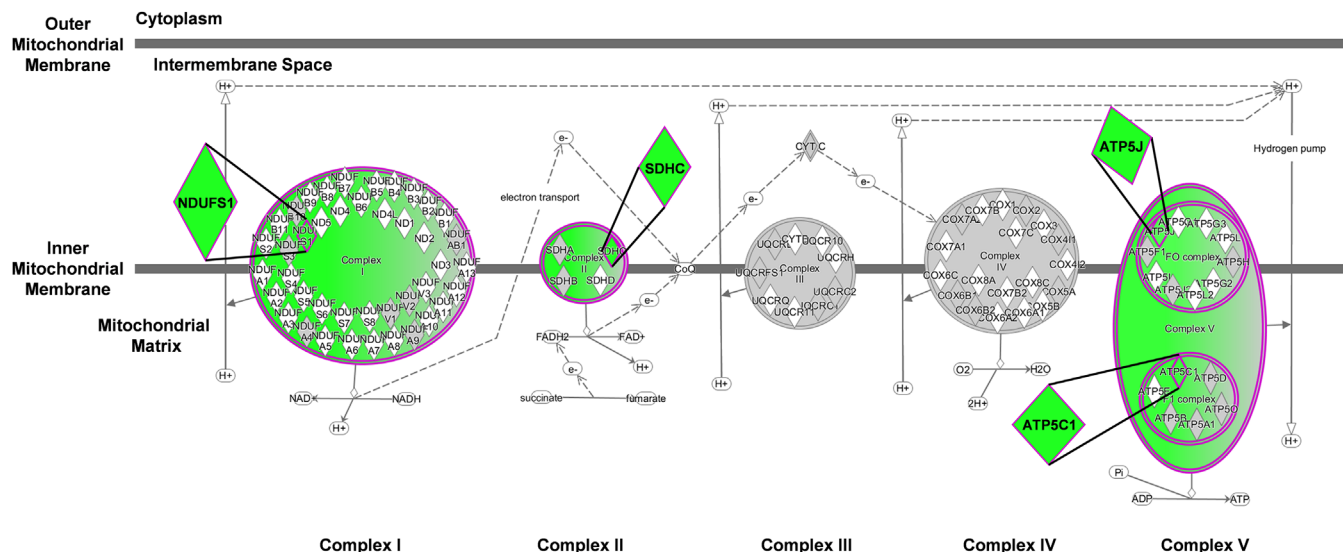


FIGURE 4 Representative differential expression of proteins in testicular tissues associated with the respiratory chain identified using Ingenuity Pathway Analysis. Red: pathways are upregulated in the TD group. Green: pathways are downregulated in the TD group. Gray: regulation was not significant

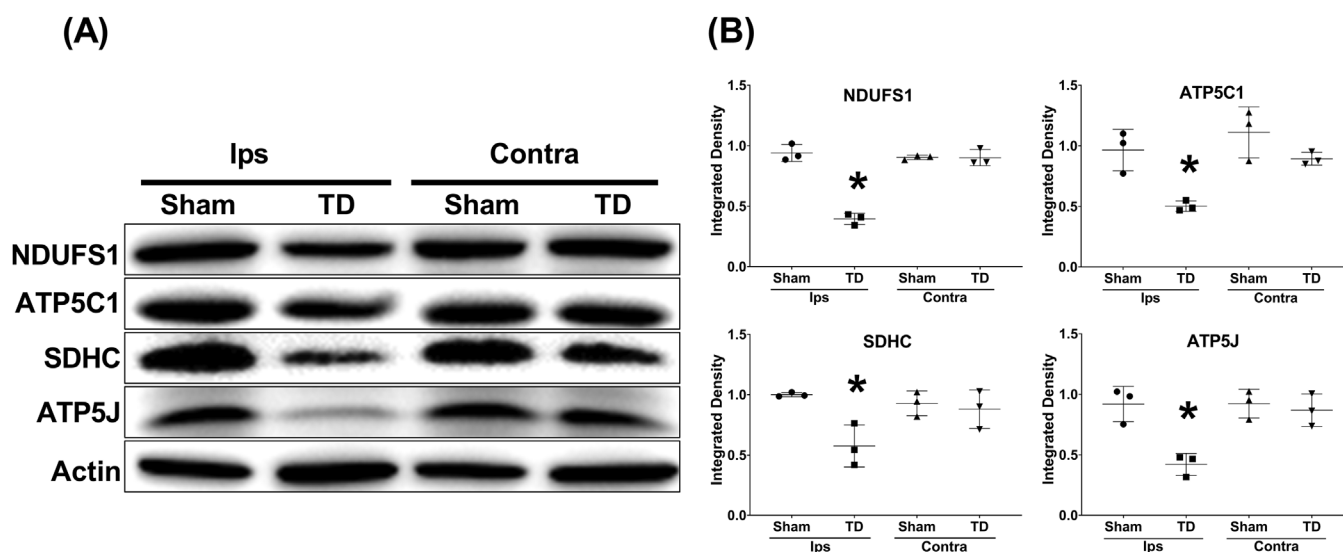


FIGURE 5 Representative results of the immunoblotting of proteins in the testicular tissues. (A) Western blot assay for the level of NDUFS1, SDHC, ATP5C1, ATP5J, and actin in testicular tissues. (B) Quantification of band intensity represented as ratio to targeting protein over actin. Data were derived from three mice in each group. Ips, ipsilateral twisted testis; Contra, contralateral untwisted testis; NDUFS1, NADH:ubiquinone oxidoreductase core subunit S1; SDHC, succinate dehydrogenase complex subunit C. * $p < 0.05$ versus Sham group

proteins. The data in Figure 5 confirmed the significantly decreased fold change of NDUFS1, SDHC, ATP5J, and ATP5C1 in ipsilateral testis of TD group compared to ipsilateral testis of Sham group.

4 | DISCUSSION

The impairment of sperm quality is a serious consequence of testicular TD.³⁻⁵ According to previous studies, intact mitochondrial function is closely related to sperm quality and fertility.^{8,9} However, little infor-

mation is available regarding mitochondrial function after testicular TD. This study demonstrated that testicular TD induced the disruption of mitochondrial morphology and decreased mitochondrial OXPHOS function through the alteration of those proteins related to OXPHOS function in the twisted ipsilateral testes but not in the untwisted contralateral testes. In a study of rats, Ahmed et al. reported that testicular torsion and detorsion did not significantly increase mitochondrial DNA deletion, and they noted no significant difference in mitochondrial oxygen consumption relative to a control group.²³ Our findings differ from those of Ahmed et al. for the following reasons. First, we used

a longer ischemia period (2 vs. 1 h) and a higher degree of torsion (720° vs. 360°). Second, we used different mitochondrial sources and different tools to evaluate mitochondrial respiratory function. Specifically, Ahmed et al. used polarographic methods with intact mitochondria isolated from testicular tissue to evaluate oxygen uptake in different groups, whereas we isolated the germ cells from testicular tissue and evaluated their mitochondrial respiratory function using a Seahorse XF Analyzer. Furthermore, we used LC-MS/MS combined with IPA to identify the expressions of proteins related to OXPHOS function. These data clearly revealed that OXPHOS dysfunction plays a crucial role in the pathogenesis of TD-induced testicular injury.

Semen quality impairment is a possible consequence of testicular TD. In a human study, a higher percentage of deformities in sperm morphology, in addition to decreased sperm motility and decreased sperm counts, were reported after testicular TD.²⁴ Several possibilities, such as pre-existing defects in spermatogenesis before torsion, reflex vasoconstriction, and antisperm antibodies, have been proposed as explanations for the relationship between testicular TD and contralateral testicular injury.^{24,25} However, the role of testicular torsion in contralateral testicular injury remains controversial. In a previous study, Shih et al. reported that the sperm motility in contralateral untwisted testes remained normal after testicular TD in rats.² Because mitochondrial function plays a major role in sperm motility, we further evaluated mitochondrial function in sperm cells after testicular detorsion. Our results revealed that the mitochondrial morphology and mitochondrial respiratory function in the TD group were not significantly different from those in the Sham group in the untwisted contralateral testes. These findings revealed that the sperm mitochondrial function in the untwisted contralateral testes was preserved after testicular detorsion.

Our results demonstrated that testicular TD reduced germ-cell respiratory function and affected those dysregulated proteins that are related to mitochondrial OXPHOS in germ cells. According to previous studies, defective mitochondrial respiratory function in germ cells results in spermatogenesis impairment and infertility.^{8,26} Proper functional mitochondrial respiratory chain complexes in sperm mitochondria play a crucial role in sperm motility.²⁷ Moreover, OXPHOS is also a major source of energy for maintaining normal function of the sperm motility.²⁸ Asthenozoospermia has been reported as a possible consequence after testicular TD, and subfertility is a serious consequence that must be prevented.²⁴ However, the role of germ cells in mitochondrial respiratory function remains unclear. Our findings clarified the status of mitochondrial respiratory function in germ cells after testicular TD, suggesting that germ-cell OXPHOS dysfunction may play a role in the pathogenesis of semen quality impairment after testicular TD.

Many medications, such as statins, FTY720, and sildenafil, have been successfully used in animal models to overcome the ischemia-reperfusion injury induced by testicular TD.^{2,29–31} However, no medication is currently available for clinical use. Proper ATP production through OXPHOS is crucial for spermatogenesis and maintaining proper semen quality.^{8,10–13} Restoration of ATP production by maintaining the respiratory function of germ cells after testicular TD may be vital for preserving semen quality and fertility. Dysregulated proteins

in OXPHOS complexes after testicular TD were discovered in this study. Novel therapeutic targets to restore dysregulated proteins in OXPHOS complexes may provide new opportunities for preserving sperm function and improving semen quality.

This study had several limitations. First, the results of this animal model cannot represent all testicular TD situations; different degrees and durations of testicular torsion induce injuries of different severities. Second, this study's findings indicated that the mitochondrial respiratory function of germ cells was altered after testicular TD. However, the question of whether preservation of mitochondrial respiratory function of germ cells can exert therapeutic effects against germ cell injuries induced by testicular TD remains unanswered. More studies are needed before further conclusions can be drawn. Third, this study only evaluated the outcome after 1 day of detorsion because urgent treatment is required after testicular detorsion to preserve testicular function. However, data from a longer follow-up course may provide additional information. Fourth, this study only evaluated the respiratory function in whole types of germ cells. Further study is needed to evaluate the respiratory function in the different type of germ cells (such as spermatogonia, spermatocyte, and spermatid).

In conclusion, testicular TD induced sperm respiratory function impairment in twisted ipsilateral testes. Dysregulated proteins in OXPHOS complexes were identified after testicular TD, and novel therapeutic targets can be identified to restore the mitochondrial OXPHOS function.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTION

All authors contributed substantially to the conception of the study as well as to data acquisition, data analysis and interpretation, and the drafting of the paper, and have approved the latest version of the submitted manuscript. All authors have agreed to be accountable for all aspects of the study and to ensure that doubts related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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SUPPORTING INFORMATION

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